

Lei Wang

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EDUCATION

University of Florida

Ph.D. Candidate in Electrical and Computer Engineering | GPA: 4.00/4.00

Gainesville, FL

Aug. 2022 – May. 2027 (Expected)

University of California, Los Angeles

M.S. in Electrical and Computer Engineering | GPA: 3.93/4.00

Los Angeles, CA

Sep. 2021 – Jun. 2022

University of Electronic Science and Technology of China

B.E. in Electronic and Information Engineering | GPA: 3.97/4.00

Chengdu, China

Sep. 2016 – Jul. 2020

PROFESSIONAL EXPERIENCES

Meta

Menlo Park, CA

Software Engineer (Machine Learning) Intern

May. 2026 – Aug. 2026

- Identified a structural blind spot in Facebook's multi-stage notification system, where the upstream timing stage is optimized independently of downstream ranking and roughly half of ~1B daily send decisions yield no notification at all, and derived a value-model reformulation, allowing a new cross-stage signal to be integrated.
- Built the cross-stage training pipeline joining timing-stage logs with downstream ranking outcomes, diagnosed and corrected label selection bias by validating the constructed label against raw source logs to within 1.5%, and trained a schedule-level multi-task model achieving 22% normalized entropy reduction and 39% AUC loss reduction over the deployed production baseline.
- Deployed to online A/B testing with two serving-time weights that tune a single trained model across the quality-volume frontier, delivering a statistically significant DAU gain, 1.3% reduction in low-quality notification volume, and 2% downstream ranking capacity savings, and extended the approach to an analogous downstream ranking stage.

PayPal

San Jose, CA

Machine Learning Scientist Intern

May. 2025 – Aug. 2025

- Identified critical limitations in production ACH fraud detection, where single-transaction models failed to predict return fraud in loyal customer segments, particularly NSF (Non-Sufficient Funds) cases with sudden high-impact loss spikes despite strong historical performance, and developed a novel contextual-temporal two-stage sequence model combining a DistilBERT foundation model for semantic feature understanding with transformer-based temporal pattern recognition to capture behavioral shift and fraud trend evolution across recent transactions.
- Validated early fraud detection capability through individual driver analysis, demonstrating the model correctly identified fraud pattern development before actual losses occurred, while production ACH risk models showed erratic signals that failed to capture sequential fraud evolution in loyal customer segments.
- Achieved significant performance improvements on 1.52M+ production ACH transactions across 30K ACH accounts: 92% relative improvement in fraud catch rates at 5% action rate, 31% reduction in false positive rates, and prevented \$1.25M additional net losses with \$3.12M gross loss reduction under \$35M Total Payment Volume compared to production ACH risk models, demonstrating superior loss prevention across all operational thresholds.

RESEARCH EXPERIENCES

FedTreeLoRA: Reconciling Statistical and Functional Heterogeneity in Federated LoRA Fine-Tuning *ICML 2026*

- Proposed **FedTreeLoRA**, a tree-structured federated LoRA fine-tuning framework that couples client-side statistical heterogeneity with layer-wise functional heterogeneity, moving beyond the flat, monolithic-adaptor assumption of prior personalized methods.
- Designed a global client hierarchy via Agglomerative Hierarchical Clustering on LoRA B-matrix distances, an adaptive layer-wise depth search to determine per-layer sharing granularity, and a Cluster-External Expert mechanism that blends peer-cluster and global knowledge through a learnable per-layer coefficient.
- Achieved state-of-the-art performance on GLUE and FLAN benchmarks with RoBERTa-Large and LLaMA-2-7B models, demonstrating up to 5.57% improvement over strongest baselines while adding negligible parameter overhead.

Adaptive LoRA Experts Allocation and Selection for Federated Fine-Tuning

NeurIPS 2025

- Proposed **FedLEASE**, a novel framework addressing two critical challenges in federated LoRA fine-tuning: determining optimal number and allocation of LoRA experts across heterogeneous clients, and enabling adaptive expert selection based on client-specific data characteristics.
- Implemented data-driven clustering approach using silhouette coefficient and cosine similarity of LoRA matrices to identify optimal expert allocation, combined with innovative adaptive top-M Mixture-of-Experts(MoE) mechanism for dynamic expert selection.
- Achieved significant performance improvements on GLUE and FLAN benchmarks with RoBERTa-Large and LLaMA-2-7B models, demonstrating 5.53% average improvement over strongest baselines while maintaining communication efficiency in heterogeneous federated settings.

Taming Cross-Domain Representation Variance in Federated Prototype Learning

NeurIPS 2024

- Designed **FedPLVM**, a dual-level prototype clustering algorithm with a novel α -sparsity prototype loss to address performance gaps in federated prototype learning across clients with heterogeneous data domains.
- Implemented the algorithm in Python using PyTorch, employing FINCH and cosine similarity metrics to ensure efficient training and model deployment across multiple environments.
- Achieved up to 9.88% higher accuracy on domain-shift benchmarks such as Digit-5, Office-10, and DomainNet, outperforming SOTA methods while optimizing communication efficiency and enhancing data privacy.

TECHNICAL SKILLS

General Skills: Machine Learning, Federated Learning, Computer Vision, Data Structure, LLM Fine-tuning

Languages/Frameworks: Python, Java, MATLAB, SQL, PyTorch, Keras, TensorFlow, Linux